



FirstEnergy Nuclear Operating Company

Perry Nuclear Power Plant
P.O. Box 97
10 Center Road
Perry, Ohio 44081

David B. Hamilton
Vice President

440-280-5382

November 30, 2018
L-18-257

10CFR50.73(a)(2)(ii)(B)

ATTN: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

SUBJECT:
Perry Nuclear Power Plant
Docket No. 50-440, License No. NPF-58
Licensee Event Report Submittal

Enclosed is Licensee Event Report (LER) 2018-003, "Unanalyzed Condition Resulting from Unfused Alternating Current Control Circuits". There are no regulatory commitments contained in this submittal.

If there are any questions or if additional information is required, please contact Mr. Glendon Burnham, Manager – Regulatory Compliance, at (440) 280-5415.

Sincerely,

David B. Hamilton
Vice President

Enclosure:
LER 2018-003

cc: NRC Project Manager
NRC Resident Inspector
NRC Region III Regional Administrator

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

Perry Nuclear Power Plant

2. Docket Number

05000-440

3. Page

1 OF 3

4. Title:

Unanalyzed Condition Resulting from Unfused Alternating Current Control Circuits

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
10	04	2018	2018	003	00	11	30	2018	Facility Name	Docket Number
										05000

9. Operating Mode

1

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

- | | | | |
|---|--|--|---|
| ☐ 20.2201(b) | ☐ 20.2203(a)(3)(i) | ☐ 50.73(a)(2)(ii)(A) | ☐ 50.73(a)(2)(viii)(A) |
| ☐ 20.2201(d) | ☐ 20.2203(a)(3)(ii) | ☒ 50.73(a)(2)(ii)(B) | ☐ 50.73(a)(2)(viii)(B) |
| ☐ 20.2203(a)(1) | ☐ 20.2203(a)(4) | ☐ 50.73(a)(2)(iii) | ☐ 50.73(a)(2)(ix)(A) |
| ☐ 20.2203(a)(2)(i) | ☐ 50.36(c)(1)(i)(A) | ☐ 50.73(a)(2)(iv)(A) | ☐ 50.73(a)(2)(x) |

10. Power Level

100

- | | | | |
|---|---|--|--|
| ☐ 20.2203(a)(2)(ii) | ☐ 50.36(c)(1)(ii)(A) | ☐ 50.73(a)(2)(v)(A) | ☐ 73.71(a)(4) |
| ☐ 20.2203(a)(2)(iii) | ☐ 50.36(c)(2) | ☐ 50.73(a)(2)(v)(B) | ☐ 73.71(a)(5) |
| ☐ 20.2203(a)(2)(iv) | ☐ 50.46(a)(3)(ii) | ☐ 50.73(a)(2)(v)(C) | ☐ 73.77(a)(1) |
| ☐ 20.2203(a)(2)(v) | ☐ 50.73(a)(2)(i)(A) | ☐ 50.73(a)(2)(v)(D) | ☐ 73.77(a)(2)(i) |
| ☐ 20.2203(a)(2)(vi) | ☐ 50.73(a)(2)(i)(B) | ☐ 50.73(a)(2)(vii) | ☐ 73.77(a)(2)(ii) |
| | ☐ 50.73(a)(2)(i)(C) | ☐ Other (Specify in Abstract below or in NRC Form 366A) | |

12. Licensee Contact for this LER**Licensee Contact**

George Dujanovic – Regulatory Compliance

Telephone Number (Include Area Code)

440-280-5200

13. Complete One Line for Each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
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14. Supplemental Report Expected☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No**15. Expected Submission Date**

Month	Day	Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

During a recent NRC inspection, it was discovered that Perry Nuclear Power Plant (PNPP) has an unfused circuit that crosses multiple safety-related fire zones. The resulting extent of condition review found an additional two components with unfused circuits also crossing multiple safety-related fire zones. Without overcurrent protection for these circuits, the potential exists that an initial fire event affecting these circuits could cause short circuits that would cause excessive current through the circuits beyond the capacity rating of the conductors. This could lead to a secondary fire in another plant area where the circuits are routed, challenging the ability to achieve and maintain safe shutdown. Compensatory fire watches have been established for the areas affected. Planned corrective actions include plant modifications to install overcurrent protection in the affected circuits.

These events are being reported under 10CFR50.73(a)(2)(ii)(B) as an unanalyzed condition. Notifications were made under 10CFR50.72(b)(3)(ii)(B), Unanalyzed Condition, on October 4, 2018 (EN 53644) and October 19, 2018 (EN 53675).

The cause of the event was due to a legacy issue stemming from the design and installation of the systems, prior to initial plant startup in 1986, that led to the vulnerability to cause a secondary fire.

The postulated events are considered to be of very low safety significance.



**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Perry Nuclear Power Plant	05000-440	YEAR 2018	SEQUENTIAL NUMBER - 003	REV NO. - 00

NARRATIVE

Energy Industry Identification System (EIS) codes are identified in the text as [XX].

INTRODUCTION

During a recent NRC inspection, it was discovered that Perry Nuclear Power Plant (PNPP) has an unfused circuit that crosses multiple safety-related fire zones. The resulting extent of condition review found an additional two components with unfused circuits also crossing multiple safety-related fire zones. Without overcurrent protection for these circuits, the potential exists that an initial fire event affecting these circuits could cause short circuits that would cause excessive current through the circuits beyond the capacity rating of the conductors. This could lead to a secondary fire in another plant area where the circuits are routed, challenging the ability to achieve and maintain safe shutdown. Compensatory fire watches have been established for the areas affected. Planned corrective actions include plant modifications to install overcurrent protection in the affected circuits.

EVENT DESCRIPTION

On October 4, 2018, with PNPP in MODE 1 at 100% rated thermal power and during the 2018 NRC Fire Protection Triennial Inspection, it was determined that the potential exists for a secondary fire to occur due to an unfused alternating current (AC) circuit [IC] associated with the Division 1 Diesel Generator 1A Ventilation Fan [VJ]. During a focused extent of condition investigation of safety-related Heating, Ventilation and Air Conditioning (HVAC) drawings on October 19, 2018, two additional unfused circuits were identified; 0M23C0002A, Motor Control Center (MCC) Switchgear and Miscellaneous Area HVAC Recirculation Fan [VI], and 0M24C0001A, Battery Room Exhaust Fan [VI]. Based on the additional findings, an expanded scope, that included a sample of non-HVAC safety-related 480V MCC fed circuits, was performed with no additional unfused circuits found.

Without overcurrent protection for these circuits, the potential exists that an initial fire event affecting these circuits could cause short circuits that would cause excessive current through the circuits beyond the capacity rating of the conductors. This could lead to a secondary fire in another plant area where the circuits are routed, challenging the ability to achieve and maintain safe plant shutdown.

Compensatory fire watches were implemented for the affected fire areas.

These conditions are being reported under 10CFR50.73(a)(2)(ii)(B) as unanalyzed conditions. Notifications were made under 10CFR50.72(b)(3)(ii)(B), Unanalyzed Condition, on October 4, 2018 (EN 53644) and October 19, 2018 (EN 53675).

CAUSE OF EVENT

The cause of the event was due to a legacy issue stemming from the design and installation of the systems, prior to initial plant startup in 1986, that led to the vulnerability to cause a secondary fire.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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		YEAR	SEQUENTIAL NUMBER	REV NO.
Perry Nuclear Power Plant	05000-440	2018	- 003	- 00

NARRATIVE**EVENT ANALYSIS**

Plant design in regard to IEEE cable separation, divisional independence, fire detection and suppression capabilities, coupled with Appendix R circuitry/method protection and Fire Protection procedural guidance provides reasonable assurance that post-fire safe shutdown will be achieved and maintained for the identified conditions. Additionally, the suspect AC control cables are insulated in cross-linked thermosetting polyethylene. Thermoset insulation as opposed to thermoplastic insulation can withstand higher temperatures during longer periods of time and burns at lower rates. Based on the expectation that any secondary fire potential would be very localized and limited (low voltage and contained to the affected cable/raceway), in addition to the aspects of the plant design and operation, there is a reasonable assurance that post-fire safe shutdown will be achieved and maintained given the postulated failure mode and the events that would have to occur in order to achieve the postulated condition.

The overall risk of this condition is judged to be very small, based on the consequential impact relative to the aforementioned aspects of plant design and the expected response to a fire event. Correspondingly, as at least one method of safe shutdown would remain available for any potential fire, this condition would be expected to be of very low safety significance.

CORRECTIVE ACTIONS:

Compensatory fire watches have been implemented for the affected fire areas. Planned corrective actions include plant modifications to install over-current protection in the respective equipment control circuitry.

PREVIOUS SIMILAR EVENTS:

On October 17, 2013, at 1300 hours with the plant in mode 1, a review of industry operating experience regarding the impact of unfused Direct Current ammeter circuits in the Control Room determined the described condition to be applicable to the Perry Nuclear Power Plant resulting in a potentially unanalyzed condition with respect to 10 CFR 50 Appendix R analysis requirements. The original plant wiring design and associated safe shutdown analysis for the Class 1E batteries control room ampere indications did not include overcurrent protection features to limit the fault current. This event was reported under 50.73(a)(2)(ii)(B) as an unanalyzed condition.

On June 19, 2014, with the plant in mode 1, a review of industry operating experience regarding the impact of unfused Direct Current (DC) circuits determined the described condition to be applicable to the Perry Nuclear Power Plant related to the Turbine Emergency Bearing Oil Pump, Reactor Feed Pump Turbine 'A' Emergency Lube Oil Pump, Turbine Emergency Seal Oil Pump, and the Reactor Feed Pump Turbine 'B' Emergency Lube Oil Pump circuits. The cause was determined to be latent design deficiency in the identified circuits dating back to original design and installation that was not identified during the design review for 10 CFR 50 Appendix R Section III.G. This event was reported under 50.73(a)(2)(ii)(B) as an unanalyzed condition.

COMMITMENTS

None